Computational Materials Science 196 (2021) 110493

Contents lists available at ScienceDirect

Computational Materials Science

journal homepage: www.elsevier.com/locate/commatsci

Investigating mechanical properties and thermal conductivity of 2D carbon-based materials by computational experiments

Luiz Felipe C. Pereira

Departamento de Física, Universidade Federal de Pernambuco, Recife 50670-901, Brazil

A B S T R A C T

Carbon is arguably one of the most versatile elements of the periodic table. Its chemical adaptability, which is due to the possible hybridization states of its electronic orbitals, allows the formation of materials with different dimensionalities. Some of the most investigated carbon-based materials: nanotubes, graphene and diamond, are know for their superlative physical properties. For instance, all three of the aforementioned materials present remarkable mechanical strength and high thermal conductivities. In fact, the relationship between mechanical strength and thermal conductivity is not unique to carbon-based materials, and tough materials tend to be good heat conductors. Nonetheless, the relationship is not always observed, and much can be learned from the materials that fall outside this behavior. Here, I present a short review of some of our results concerning the mechanical properties and the lattice thermal conductivity of two-dimensional carbon-based materials, obtained by state-of-the-art computational experiments.

1. Introduction

The chemical versatility of carbon atoms allows them to exist with sp, sp^2 and sp^3 orbital hybridizations, which reflect in diverse all-carbon structures. In bulk, diamond is the most stable carbon allotrope with its complete sp^3 hybridized atoms. In lower dimensions, carbon atoms form fullerenes [1], carbon nanotubes [2] and graphene [3–5], all of which contain exclusively sp^2 hybridized atoms. For many decades diamond was the reining carbon allotrope in terms of mechanical strength and thermal conductivity, until the advent of carbon nanotubes and later graphene. The now ubiquitous two-dimensional (2D) carbon allotrope came about from early experiments by Geim and Novoselov and rapidly became the rising star on the horizon of materials science and condensed-matter physics [4].

In 2004, Novoselov and coworkers reported the observation of the electric field effect in atomically thin monocrystalline graphitic films, including few-layer and single-layer graphene, prepared by mechanical exfoliation [3]. Their work is regarded as the first experimental report on the physical properties of graphene, a material which had been presumed not to exist freely, since it would be unstable with respect to curved carbon structures such as soot, fullerenes, and nanotubes. In the following year Novoselov and coworkers reported the production of free-standing 2D atomic crystals, which can be viewed as individual atomic planes pulled out of bulk crystals. They employed micromechanical cleavage to prepare a variety of high quality 2D crystals including single layers of boron nitride, graphite, several dichalcogenides, and complex oxides, all of which are stable under ambient conditions [6]. Soon after the seminal reports by Novoselov and coworkers the interest in graphene and other 2D materials grew remarkably fast.

In 2008, Hernandez and coworkers developed a high-yield method to produce graphene by liquid-phase exfoliation of graphite, and in 2011, Coleman and coworkers extended the liquid exfoliation method to obtain nanosheets of boron nitride and several dichalcogenides [7,8]. In 2009, Li and coworkers grew large-area single-layer graphene films on copper substrates by chemical vapor deposition using methane, thus enabling studies with graphene samples of the order of centimeters [9]. The availability of high-quality samples of 2D materials further increased the interest in their physical properties, including their elastic modulus, their mechanical strength and their lattice thermal conductivity [4,10–14].

Here, I present a short review of mechanical and heat transport properties of single-layer carbon-based materials. I begin by presenting some of the computational methods employed to calculate the mechanical properties and thermal transport characteristics of materials. I then review some of the early measurements of the elastic properties and lattice thermal conductivity of single-layer graphene, followed by the respective predictions by computational methods. Afterwards, I summarize our contribution to the determination of the lattice thermal conductivity of graphene via molecular dynamics simulations, including some of the difficulties found along the way. Following this initial summary, I review our computational results for several carbon-based 2D materials, and conclude this short review presenting a general perspective for the near future.

E-mail address: pereira@df.ufpe.br.

https://doi.org/10.1016/j.commatsci.2021.110493
Received 30 December 2020; Received in revised form 26 March 2021; Accepted 1 April 2021
Available online 5 May 2021

2. Computational methods

For hundreds of years humankind employed two main approaches to investigate, describe and explain the behavior of nature and its surroundings: experiments (or observations) and theories (or hypotheses). In general, one can say that an observation leads to a hypothesis, which can be formalized in a theory, which then predicts the result of an experiment to be performed, thus validating or nullifying the theory. The advent of modern computers provided an alternative way to probe nature: computer simulations. In fact, as soon as large scale computers became available the first computer experiments were performed by Metropolis and coworkers [15], as well as Fermi, Pasta and Ulam [16].

Molecular dynamics (MD) simulation is one of the techniques that can be used to perform computational experiments. Indeed, MD is a powerful computational tool that can be used to investigate materials in a broad range of pressure and temperature conditions. The basic procedure of MD consists of integrating Newton's equations of motion in discrete time steps for each atom subject to forces applied by neighboring atoms. Interatomic forces can be calculated from empirical potentials, which are usually parameterized by experimental measurements [17], or from the self-consistent solution of the electronic structure of the system by first principles methods [18]. Regardless of how the interatomic forces are calculated, the dynamics of the system is inherently classical since it is determined by Newton's equations of motion. MD simulation based on empirical potentials can be applied to systems with more than a million atoms, while first principles methods are more computationally demanding and limited to systems with hundreds of atoms.

2.1. Mechanical properties

The mechanical properties are a fundamental aspect to be considered in the search for new materials and their possible applications. Much information about the mechanical properties of a material can be obtained from its stress–strain relation. In MD simulations, the Cartesian components of the stress tensor are evaluated from atomic positions, velocities and forces, and can be calculated by

$$\sigma_{\mu\nu} = -\frac{1}{V}\left(\sum_i m_i v_{i,\mu} v_{i,\nu} + \sum_i r_{i,\mu} F_{i,\nu}\right), \tag{1}$$

where V is the system volume, m_i is the mass of atom i, $v_{i,\mu}$ is a component of the velocity of atom i, $r_{i,\mu}$ is a component of the position and $F_{i,\nu}$ is a component of the force acting on atom i. The stress tensor is symmetric, thus only 6 of its components are independent: the three diagonal terms and three of the remaining off-diagonal terms. The diagonal terms relate to compressive or tensile deformations, while the off-diagonal terms are related to shear deformations. In the case of 2D materials, in-plane tensile deformations are the most relevant, and the volume is defined as the product of the surface area with a nominal thickness. Therefore, it is sometimes convenient to replace the volume by the surface area and define a 2D stress in units of pressure × length or force/length, instead of the usual units of pressure given by force/area.

Strain is a measure of the deformation representing a displacement in the body relative to its reference configuration. The strain tensor is directly related to the stress tensor by the constitutive equations derived from Hooke's law [19]

$$\sigma_{\mu\nu} = C_{\mu\nu\alpha\beta}\epsilon_{\alpha\beta}, \tag{2}$$

where $C_{\mu\nu\alpha\beta}$ is the stiffness tensor or elastic tensor. Much like the stress tensor, the strain tensor is also symmetric with 6 independent components: the three diagonal terms (representing compressive or tensile deformations) and three of the remaining off-diagonal ones (representing shear deformations). Therefore, the elastic tensor has at most 36 independent elements, although symmetry considerations can reduce these to a minimum of 4 independent elements. In the case of 2D materials, the most relevant elements of the elastic tensor are $C_{xxxx} = C_{11}$, $C_{yyyy} = C_{22}$, which correspond to the Young's modulus along both in-plane directions.

The uniaxial engineering strain is defined as the relative change in length along one specific direction, such that

$$\epsilon_{\mu\mu} = \frac{L_\mu - L_\mu^0}{L_\mu^0} = \left(\frac{L_\mu}{L_\mu^0} - 1\right), \tag{3}$$

where L_μ^0 is the original length along that direction, and L_μ is the length after deformation. According to this definition, a tensile strain corresponds to positive values and a compressive strain corresponds to negative values. Another important elastic property of a material is the Poisson ratio, defined as the ratio between the strain transverse to the deformation direction and the strain along the deformation direction

$$\nu = -\frac{\epsilon_{\alpha\alpha}}{\epsilon_{\beta\beta}}. \tag{4}$$

In the case of a 2D material, α and β are perpendicular in-plane directions. Therefore, the Poisson ratio measures the contraction ($\nu > 0$) or expansion ($\nu < 0$) of a 2D material perpendicular to its tensile deformation direction.

In order to determine stress–strain curves, the simulated samples are subject to uniaxial or biaxial tensile strain through gradual elongation of one or two in-plane directions simultaneously. Sample deformation can be achieved by a stepwise displacement of the atoms in the simulation cell, such that their final coordinate along the strained direction is given by $\boldsymbol{r}_i = (1 + \epsilon_{\mu\mu})\boldsymbol{r}_i^{\mathbf{0}}$, where r_i^0 is the initial position of atom i, and $\epsilon_{\mu\mu}$ is the applied engineering strain. In general the stress–strain curves present an initial linear region, which corresponds to the elastic deformation of the material and from where the in-plane Young's moduli and Poisson ratio are calculated, followed by a non-linear region, which corresponds to the plastic deformation and the mechanical failure of the material. Usually, the rupture of the material is characterized by a quick drop in the stress, and the maximum stress before rupture (ultimate stress) defines the tensile strength of the material, to which the ultimate strain is associated.

One important aspect of the this type of simulation is the need to control the rate at which the system is strained. If the deformation happens too fast it can lead to unphysical results in the stress–strain diagram, whereas if it is too slow the simulation can take an unnecessarily long time. Typically, strain rates of the order of 10^8 s^{-1} are employed. Also, in the case of uniaxial strain, it is also important to let the system length perpendicular to the strained direction relax, in order to capture the mechanics of the materials and evaluate its Poisson ratio. An extensive review including theoretical and experimental works related to the mechanical properties of 2D materials has recently been presented by Akinwande and coworkers [20].

2.2. Lattice thermal conductivity

Molecular dynamics simulations offer different methods to calculate the thermal conductivity of a system. In a broad sense, those methods can be divided in two categories: equilibrium and non-equilibrium methods. In the equilibrium method (EMD), the system is kept at a constant homogeneous temperature and the thermal conductivity is obtained from fluctuations of the heat current, in the form of its autocorrelation function. Following linear response theory, each component of the thermal conductivity tensor can be expressed according to the Green–Kubo formula [21,22]

$$\kappa_{\mu\nu}(t) = \frac{1}{Vk_BT^2}\int_0^t \langle J_\mu(0)J_\nu(t')\rangle dt' \tag{5}$$

where k_B is Boltzmann's constant, V is the volume of the system, T is the absolute temperature, $J_\mu(t')$ is the component of the heat current along Cartesian direction μ, and t is the correlation time. In practice, κ is taken as the stationary value of Eq. (5) before it drifts due to accumulated statistical noise. Although κ is a tensor, for 2D materials we have $\kappa_{xx} = \kappa_{yy}$ and $\kappa_{zz} = 0$. Also, all the non-diagonal terms vanish.

The heat current is defined in terms of atomic positions and energies in the from

$$\boldsymbol{J} = \frac{d}{dt}\sum_i \boldsymbol{r}_i E_i, \tag{6}$$

where $E_i = m_i v_i^2/2 + U_i$ is the total energy of atom i, m_i is the atomic mass, $\boldsymbol{v}_i$ is the velocity and U_i is the potential energy. Therefore, it is possible to write the heat current in terms of atomic quantities and its derivatives as

$$\boldsymbol{J} = \sum_i \boldsymbol{v}_i E_i + \sum_i \boldsymbol{r}_i(\boldsymbol{F}_i \cdot \boldsymbol{v}_i) + \sum_i \boldsymbol{r}_i \frac{dU_i}{dt}, \tag{7}$$

where $\boldsymbol{F}_i$ is the force on acting on atom i. The first term is the so-called kinetic or convective term, and in the case of solid systems it is negligible. The remaining terms are the potential terms, and those carry the most importance in the case of solid materials.

Notice that it was not necessary to assume any particular form for the potential energy U_i in the above derivation. However, the definition of U_i amounts to a decomposition of the total potential energy of the many particle system into atomic potential energies. Such a decomposition is not needed for the derivation of the atomic forces, but it could be crucial for the correct derivation of the heat current, which involves a time derivative of U_i. There is no clear physical intuition favoring one particular decomposition over others, but one must be consistent with the chosen definition. In fact, recent developments have shown that the transport coefficients are essentially independent of the microscopic representation chosen for the current density [23], and this gauge invariance has been used to derive a general approach to describe the thermal conductivity of crystalline solids and glasses [24]. Independently, other approaches have recently been developed to describe the thermal conductivity of semiconductors and insulators [25], as well as anharmonic crystals and harmonic glasses [26].

Another way of calculating the thermal conductivity within MD simulations is to drive the system out of equilibrium, in the so-called non-equilibrium methods (NEMD). In the traditional or direct non-equilibrium method, a temperature gradient is imposed in the system originating a heat flux which can be measured [27]. The temperature gradient can be imposed and sustained by thermostats connected to separate regions of the system, which exchange energy with the atoms in each region. Meanwhile, in the reverse non-equilibrium method, a heat flux is imposed in the system originating a temperature gradient which can be measured [28]. The heat flux can be imposed and sustained by exchanging the kinetic energy of slow moving atoms in one region of the system with fast moving ones in another region of the same system.

Regardless of which particular variant of the NEMD method is chosen, the thermal conductivity is calculated directly from the temperature gradient and the heat flux via Fourier's law $\boldsymbol{J} = -\kappa\nabla T$. In general the temperature gradient is imposed along a single Cartesian direction, and the heat flux is measured along the same direction, or vice versa, such that one can obtain a specific component of the thermal conductivity tensor

$$\kappa_{\mu\mu} = -\frac{J_\mu}{dT/d\mu}. \tag{8}$$

Although those methods consider systems out of equilibrium, it is necessary to achieve a stationary regime, where the average value of the heat flux and the temperature gradient remain approximately constant. The time necessary to reach the stationary regime depends on each specific the system, but it is generally on the order of 10^8 simulation time steps.

The NEMD methods mimic the experimental setup of most measurements, where a temperature gradient is generated in the system originating a heat flux [12]. Another variant of the NEMD method, which does not require a constant heat flux in the system, is the approach-to-equilibrium MD method [29,30]. In this method, the lattice thermal conductivity is calculated during the transient regime from an initial non-uniform temperature profile to a final equilibrium situation.

It is also possible to calculate the thermal conductivity by anharmonic lattice dynamics, which is a valuable complement to MD simulations, and relies on solutions of the Boltzmann transport equation (BTE) [31,32]. Within this formalism, the phonon properties of the material are calculated from its dynamical matrix, which is the Hessian of the interatomic potential, as well as higher order derivatives of the potential which account for the anharmonic terms. The eigenvectors of the dynamical matrix are the phonon modes of the system and its eigenvalues are the square of the phonon frequencies. The lattice thermal conductivity is calculated directly from the BTE, solved in the single mode relaxation time approximation or more accurately by iterative and variational methods [33–38].

The contribution of each phonon mode to the thermal conductivity of the system can be cast in terms of its heat capacity, group velocity and lifetime. The heat capacity and the group velocity of each phonon mode are calculated from its frequency and it dispersion relation, respectively, and correspond to purely harmonic terms. Meanwhile, the lifetime of each mode is determined by the anharmonic terms, which include the interaction with other phonon modes. In many cases, it is enough to consider 3-phonon scattering terms, but systems with higher anharmonicity require also the calculation of 4- and 5-phonon scattering terms. Finally, in order to estimate the thermal conductivity of naturally occurring structures, isotope scattering can also be considered in the solution of the BTE.

Notice that the phonon properties needed to calculate the thermal conductivity with the BTE method can be obtained from classical interatomic potentials or from first principles calculations [33,39,38]. Each method has its own advantages and disadvantages, but in general using classical methods one can treat systems with several thousand atoms while first principles methods are limited to a few hundred. Direct comparison has been made between MD-based methods and the BTE method, which show a general numerical agreement among them as long as each method is applied within its range of validity [40–43]. A comparative analysis, along with a critical description of the merits and drawbacks of MD-based methods and the BTE approach has recently been discussed in a detailed tutorial review by Fugallo and Colombo [44].

3. A brief history of the elastic properties and thermal conductivity of graphene

The elastic properties and the intrinsic breaking strength of free-standing monolayer graphene membranes were first measured by nanoindentation in an atomic force microscope by Lee and coworkers in 2008 [45]. Interpreting the force–displacement behavior within a nonlinear elastic stress–strain response framework, they found a second-order elastic stiffness of $E^{2D} = 340 \pm 50$ N m^{-1} and an intrinsic breaking strength of $\sigma^{2D} = 42 \pm 4$ N m^{-1} for a defect-free graphene sheet. Assuming that the thickness of a single-layer graphene sheet equals the interlayer separation in graphite (0.335 nm), the above quantities correspond to a Young's modulus of $E^{3D} = 1.0 \pm 0.1$ TPa and an intrinsic strength of $\sigma^{3D} = 130 \pm 10$ GPa at an ultimate strain of 0.25. For comparison, in 1970, Blakslee and coworkers determined the elastic constants of pyrolytic graphite ordered by annealing under compressive stress [46]. Their measurements yielded an in-plane Young's modulus of 1.02 ± 0.03 TPa for highly ordered graphite.

In 1998, Hernandez and coworkers used the tight-binding formalism to perform a comparative study of the elastic properties of carbon-based nanostructures [47]. They predicted $E^{2D} = 0.41$ TPa·nm for graphene, which considering a thickness of 0.335 nm, yields $E^{3D} = 1.2$ TPa. In 2001, Kudin, Scuseria and Yakobson employed first principles methods to predict the mechanical properties of 2D carbon lattices, including graphene [48]. Their calculations resulted in a 2D Young's modulus $E^{2D} = 345$ N m^{-1}, and a 3D Young's modulus $E^{3D} = 1029$ GPa. In 2007, Liu and coworkers employed density functional perturbation theory to investigate the phonon spectra of graphene under uniaxial strain along the armchair or zigzag directions [49]. They obtained a Young's modulus $E^{3D} = 1050$ GPa along either in-plane directions, with a direction-dependent failure stress of 110 GPa for the zigzag direction and 121 GPa for the armchair direction. In 2009, Cadelano and coworkers combined continuum elasticity theory and tight-binding atomistic simulations, to obtain the constitutive nonlinear stress–strain relation for graphene and calculated the corresponding nonlinear elastic moduli [50]. Their method resulted in a 2D Young's modulus of $E^{2D} = 312$ N m^{-1}, and a failure stress of $\sigma^{2D} = 42.4$ N m^{-1}.

In 2009, Jiang, Wang and Li performed MD simulations based on the second generation Brenner interatomic potential to investigate the Young's modulus of graphene [51,52]. For $T = 300$ K, their result $E^{3D} = 1050 \pm 50$ GPa was in good agreement with the experiments by Lee and coworkers [45] and the first principles results of Kudin, Scuseria and Yakobson, and Liu and coworkers [48,49]. In 2012, Mortazavi and coworkers used classical MD simulations to investigate the tensile behavior of graphene based on two versions of the Tersoff bond-order empirical potential [53–56]. The original parameterization [54,55] predicted $E^{3D} = 1050$ GPa while a parameter set optimized for graphene [56] predicted $E^{3D} = 985$ GPa, both in good agreement with previous experimental, theoretical and computational results.

In 2008, Balandin and coworkers reported the first experimental study of the thermal conductivity of single-layer graphene obtained by mechanical exfoliation and suspended over wide trenches in a Si/SiO_2 substrate [57]. Using a non-contact optical-based technique, they extracted the thermal conductivity from the dependence of the Raman G peak frequency on the power of the excitation laser. The measurements yielded an extraordinary high thermal conductivity up to $\kappa = 5300 \pm 480$ W/m-K at room temperature. In 2010, Cai and coworkers measured the thermal conductivity of suspended graphene monolayers grown by chemical vapor deposition on copper [58]. They improved the non-contact Raman technique by adding a power meter under the suspended portion of graphene, and found that the conductivity of high-quality samples exceeded 2500 W/m-K at 350 K and reached 1400 W/m-K at about 500 K. Still in 2010, Jauregui and coworkers combined Raman spectroscopy for thermometry and electrical transport for Joule heating to measure the thermal conductivity of suspended graphene grown by chemical vapor deposition, obtaining values in the range from 1500 to 5000 W/m-K [59].

More than a decade before the advent of graphene, Klemens and Pedraza derived theoretical expressions for the lattice thermal conductivity of single crystals of graphite in the basal plane, and obtained the value of 1910 W/m-K at 300 K, in good agreement with later experiments in graphene [60,58,59]. In 2009, Nika and coworkers developed a theoretical framework to determine the lattice thermal conductivity of single-layer graphene, based on a valence force field method and treating three-phonon processes exactly [61]. They found a near room temperature thermal conductivity in the range from 2000 to 5000 W/m-K. In the same year Kong and coworkers used first principles calculations to obtain a lattice thermal conductivity around 2200 W/m-K at 300 K for an ideal graphene monolayer [62]. In 2010, Evans, Hu and Keblinski used equilibrium MD simulations based on a variant of the Tersoff potential [63] to compute the thermal conductivity of graphene nanoribbons, and found that it converged to values in the range from 8000 to 10000 W/m-K as the width increased beyond 50 Å and the ribbons approached graphene sheets [64]. Meanwhile, Thomas, Iutzi and McGaughey employed NEMD simulations based on the second generation REBO potential [51] to predict a lattice thermal conductivity of 350 W/m-K [65].

This was the state of affairs in 2010 when Lindsay and Broido reparameterized the Tersoff and Brenner empirical potentials originally designed for diamond, to better represent the lattice dynamical properties of graphene, such as the dispersion of the acoustic phonons [56]. Employing the BTE approach, the room temperature lattice thermal conductivity of graphene with the original parameter sets were 1900 W/m-K for the Tersoff case and 1100 W/m-K for the Brenner case. With the optimized potentials, Lindsay and Broido obtained lattice thermal conductivities of 3500 W/m-K for the Tersoff case and 3600 w/m-K for the Brenner case. Besides the mutual agreement between both bond-order potentials, the optimized potentials also achieved good agreement with experimental measurements [12].

Nonetheless, in spite of the experimental, theoretical, and computational results available, one fundamental aspect regarding the lattice thermal conductivity of graphene was still unclear: whether its intrinsic value was ultimately upper-limited or if it was length-dependent and eventually divergent in the thermodynamic limit. In fact, the thermal conductivity of non-linear 2D lattices was investigated by Fermi, Pasta and Ulam in one of the first computational experiments ever performed [16]. In the case of 2D lattices, the thermal conductivity depends on the system size and diverges logarithmically with its length along the heat current direction [66–70]. Therefore, back in 2010, it was still unclear if graphene should behave as a 2D system in regards to its phonon heat transport or if its lattice thermal conductivity was indeed size-independent in the thermodynamic limit.

4. Computing the lattice thermal conductivity of graphene and the heat flux issue

The work of Lindsay and Broido provided a reliable parameterization of the Tersoff potential for carbon nanostructures such as graphene and carbon nanotubes [56]. However, much uncertainty remained in the computational determination of the lattice thermal conductivity of graphene. Studies employing the same empirical potential and parameter set provided a wide range of values for the lattice thermal conductivity depending on the calculation method and sometimes on the software employed. Of course, this was a very unsatisfactory situation which jeopardized the predictive power of computer simulations. In 2011, in the research group led by Davide Donadio, I started computing the lattice thermal conductivity of graphene via MD simulations with LAMMPS [71], employing the equilibrium method with the Green–Kubo formalism and non-equilibrium methods with the direct and the reverse methods.

First we considered possible size-effects on the thermal conductivity of graphene calculated with EMD simulations, as well as the effect of mechanical strain and isotopic disorder [72]. We performed simulations with approximately square supercells of increasing size, from 26 x 25 Å^2 (240 C atoms) to 880 x 877 Å^2 (3 x 10^5 atoms), calculating the conductivity along both in-plane directions independently, as shown in Fig. 1(a). The smallest system presented an anisotropy in κ, due to an uneven and insufficient sampling of the vibrational modes. Our results indicated a converged value $\kappa = 1015 \pm 120$ W/m-K for a graphene sample with 17200 atoms. We also found that κ increased with strain and eventually diverged for graphene samples under uniaxial strain above 2%. The simulations showed that the presence of isotopic mass disorder decreased κ but not enough to suppress its divergence.

In the following year we presented our results obtained from NEMD simulations, along with experimental results obtained at the National University of Singapore [73]. In this case, both the direct and the reverse methods presented a remarkable numerical agreement for the length-dependent thermal conductivity, show in Fig. 1(b), in qualitative agreement with experimental measurements. However, a large discrepancy with the values obtained in our previous EMD study

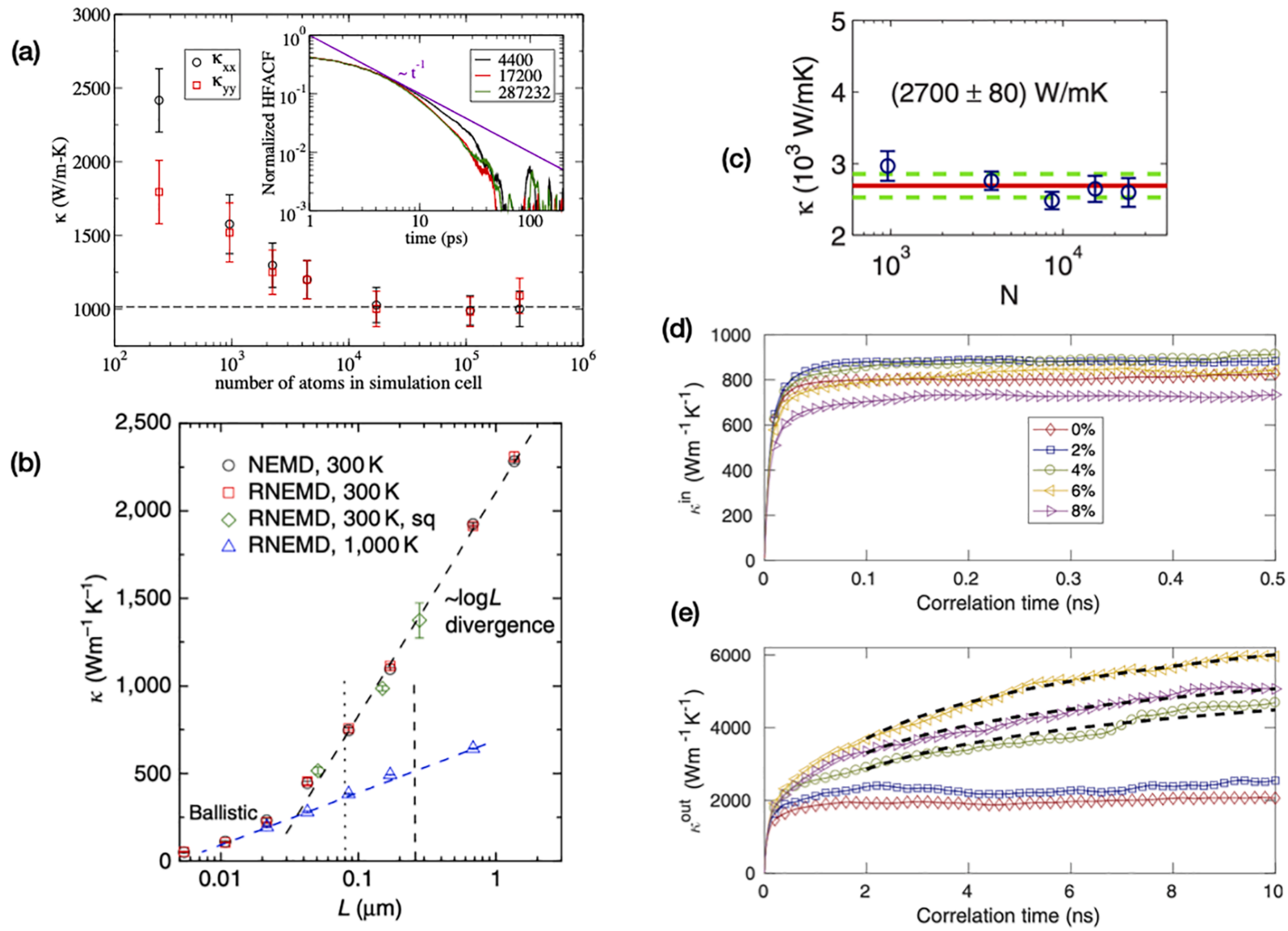

Fig. 1. (a) Lattice thermal conductivity of graphene as a function of the number of atoms in the sample. Inset: decay of the heat flux autocorrelation function. Adapted from Ref. [72]. (b) Size dependent thermal conductivity of graphene from non-equilibrium molecular dynamics simulations. Adapted from Ref. [73]. (c) Lattice thermal conductivity of graphene as a function of the number of atoms in the simulation cell, calculated with the correct many-body heat flux expression. Adapted from Ref. [75]. (d) In-plane and (e) out-of-plane contributions to the running thermal conductivity of graphene under uniaxial strain. Adapted from Ref. [76].

remained: at 300 K our NEMD simulations predicted $\kappa = 2300$ W/m-K for samples of 1.5 μm, much larger than the converged value obtained from EMD simulations. Since the NEMD results were much closer to the experiments than the EMD ones, we concluded that there must be some inconsistency in the latter.

Soon after our publication, Fugallo and coworkers developed an exact solution of the BTE with phonon–phonon collision rates obtained from first principles calculations [35]. They showed that collective phonon excitations (as opposed to single phonons) with mean free paths of the order of hundreds of micrometers are the main heat carriers in graphene and other 2D materials [35]. In the case of graphene, they concluded that sample sizes of the order of 1 mm were required in order to achieve the intrinsic thermal conductivity at room temperature. In 2015, Barbarino, Melis and Colombo performed approach-to-equilibrium MD simulations based on the second generation REBO interatomic potential to simulate graphene samples up to 0.1 mm [74]. They concluded that the thermal conductivity is indeed upper limited in samples long enough to allow diffusive transport for both single and collective phonon excitations [74]. Therefore, both independent studies demonstrated that computational experiments predict a size-independent lattice thermal conductivity for graphene, as long as the simulation cells are large enough.

In the months following those developments, Zheyong Fan and Ari Harju pointed out inconsistencies in the definition of the heat flux implemented in LAMMPS, and we began working together on understanding the nature of this issue. We were able to conclude that the expression used in LAMMPS to calculate the heat flux was only valid for two-body potentials, and as such, provided incorrect results for systems described by many-body potentials such as Tersoff, Brenner and Stilinger-Weber [75]. Furthermore, we were able to derive the appropriate expression for many-body potentials and verify its correctness for systems such as silicon, diamond, carbon nanotubes and graphene. Our calculations reproduced in Fig. 1(c) yielded an intrinsic thermal conductivity of 2700 ± 80 W/m-K for graphene at 300 K, in agreement with experiments and NEMD simulations.

In 2017, we expanded this formulation to decompose the microscopic heat current in atomistic simulations into in-plane and out-of-plane components [76]. We applied the decomposition to study heat transport in suspended graphene, using both EMD and NEMD simulations. Finally, we arrived at a combined EMD and NEMD value for the lattice thermal conductivity of graphene $\kappa = 2900 \pm 100$ W/m-K. As shown in Fig. 1(d) and (e), we concluded that the flexural component is responsible for about two-thirds of the total thermal conductivity in unstrained graphene, and the acoustic flexural component is responsible for the divergence of the conductivity when a sufficiently large tensile strain is applied, in a remarkable agreement with our initial prediction [72]. To date, this results remains to be verified experimentally.

5. Mechanical properties and thermal conductivity of 2D carbon allotropes

5.1. Amorphized graphene: a stiff material with low thermal conductivity

In 2015, Kotakoski and coworkers showed that carbon atoms in pristine single-layer graphene could be removed or rearranged into a random pattern of polygons using a focused beam of Ga^{+} ions [77]. Atomic resolution scanning transmission electron microscopy (STEM) showed that the amorphous areas remained two-dimensional, consisting of a variety of carbon rings with sizes ranging from five to nine atoms, with occasional pores, similar to amorphous 2D carbon structures. Inspired by their experiments, we decided to investigate the mechanical properties and the thermal conductivity of amorphized graphene sheets, illustrated in Fig. 2(a) via MD simulations based on the optimized Tersoff potential reparameterized by Lindsay and Broido [78]. The defect concentration was defined as the ratio between the number of non-hexagonal rings in the amorphized structure and the total number of hexagons in the initial pristine sheet.

We investigated the mechanical properties of pristine and amorphized graphene by imposing a uniaxial tensile strain at different temperatures: 300 K, 500 K and 700 K. During loading, the length of the simulation box along the loading direction was increased at every time step by a constant engineering strain rate of 2×10^{8} s^{-1}, while the pressure along the perpendicular direction was coupled to a barostat in order to keep it around zero, on average. The elastic modulus of pristine graphene was calculated from the slope of the initial linear region of the stress–strain curves, which corresponded to an elastic modulus $E^{3D} = 960\pm 10$ GPa and a tensile strength $\sigma^{3D} = 132$ GPa at 300 K, in excellent agreement with the experimental results of Lee and coworkers [45]. In the case of amorphized graphene, we observed a decrease in elastic modulus as the concentration of defects increased, going from $E^{3D} = 850$ GPa at 5% down to $E^{3D} = 500$ GPa at 35% defect concentration. Tensile strength also decreased from $\sigma^{3D} = 95$ GPa at 5% down to $\sigma^{3D} = 50$ GPa at 35% defect concentration, while the ultimate strain increased from 0.12 at 5% to 0.16 at 35% defect concentration, as shown in Fig. 2 (b). For pristine and amorphized graphene both tensile strength and ultimate strain decreased with temperature in the range from 300 K to 700 K, which was expected and consistent with general knowledge.

The thermal conductivity of pristine and amorphized graphene were investigated with the EMD method, since NEMD produced instabilities in the defective samples, which were more pronounced for higher defect concentrations and temperatures. Nonetheless, we employed the correct formulation for the heat flux derived by ourselves in 2015 [75], which predicted a thermal conductivity of 2700 ± 80 W/mK for pristine graphene at 300 K, in agreement with NEMD studies employing the same parameters for the Tersoff potential [73,79,80]. The dependence of the thermal conductivity with defect concentration is presented in Fig. 2(c) for temperatures of 300 and 500 K. In the case of amorphized samples at 300 K, we obtained $\kappa = 95\pm 10$ W/m-K for 5% and $\kappa = 15\pm 2$ W/m-K at 35% defect concentration. This significant reduction in thermal conductivity reflects the reduction in the phonon group velocities,

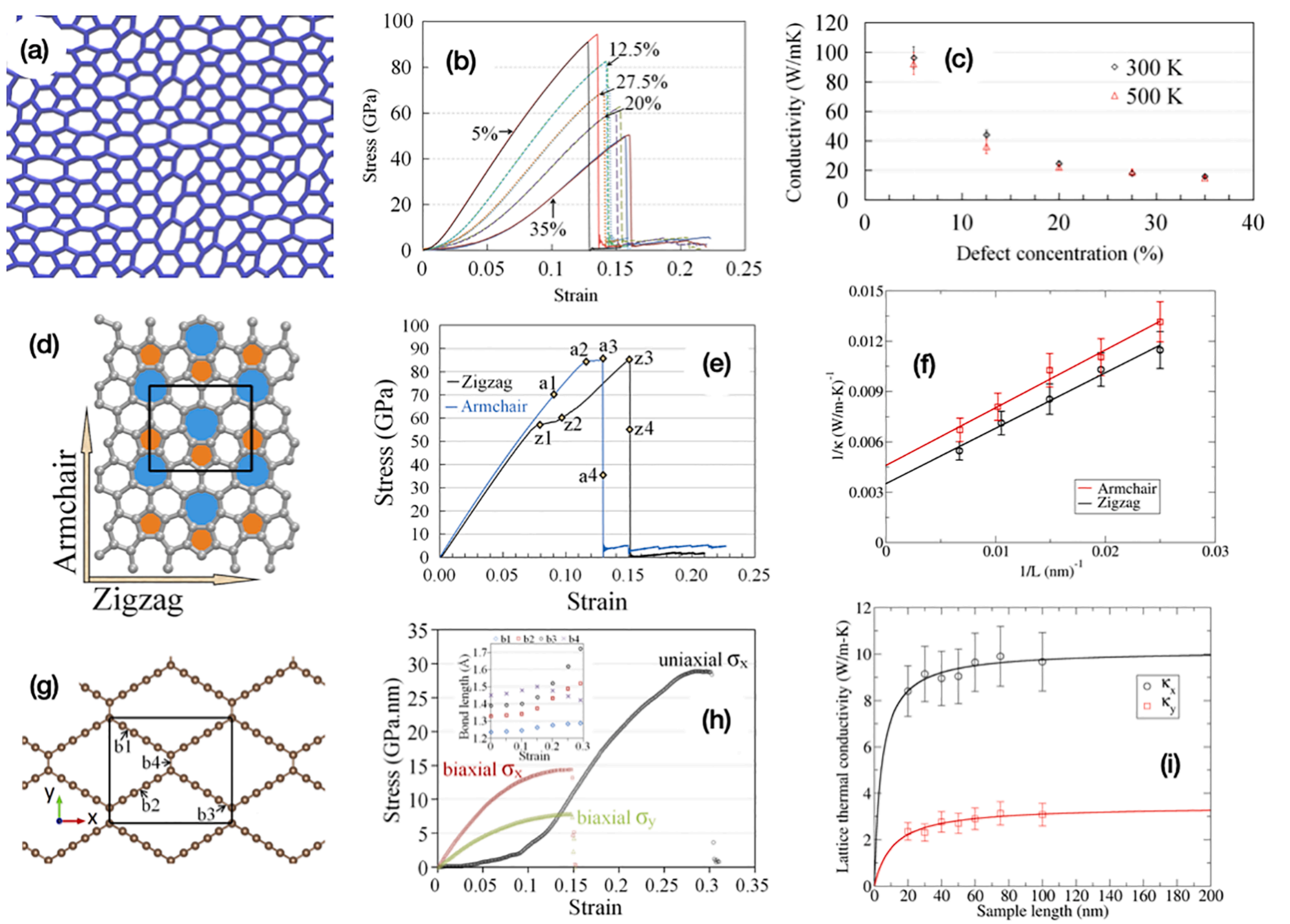

Fig. 2. (a) to (c) Atomic structure, stress–strain curve and Green–Kubo thermal conductivity of amorphized graphene. Adapted from Ref. [78]. (d) to (f) Atomic structure, stress–strain curve and inverse thermal conductivity of phagraphene. Adapted from Ref. [83]. (g) to (i) Atomic structure, stress–strain curve and thermal conductivity of 2D ene-yne graphyne. Adapted from Ref. [84].

which are much smaller in the case of amorphized graphene. Combining the mechanical and heat conduction properties, we concluded that amorphized graphene presents a remarkably high elastic modulus and mechanical strength, along with a very low thermal conductivity, which is an unusual combination for carbon-based materials. This unique feature could be useful to produce thermally insulating composites with high mechanical resistance.

A few months after our publication, Malekpour and coworkers reported their measurements of the thermal conductivity of graphene with defects induced by low-energy electron beam irradiation [81]. The thermal conductivity of suspended graphene was measured with the optothermal Raman technique, and compared to solutions of the BTE as well as MD simulations with the optimized Tersoff potential. Near room temperature, Malekpour and coworkers observed a decrease from $\kappa \approx (1.8 \pm 0.2) \times 10^3$ W/m-K to $\kappa \approx (4.0 \pm 0.2) \times 10^2$ W/m-K, as the defect density increased from 2.0×10^{10} cm^{-2} to 1.8×10^{11} cm^{-2}. Their measurements indicated a saturation-type behavior at a relatively high value of $\kappa \approx 400$ W/m-K at higher defect densities, and their MD simulations also indicated a saturation at $\kappa \approx 800$ W/m-K. In their simulations, Malekpour and coworkers considered as defects only single and double vacancies, due to the energy scales involved in the electron irradiation process. Meanwhile, our simulations introduced defects in the form of non–hexagonal rings, mainly penta-, hepta- and octagonal rings, with eventual vacancies appearing as a by-product of these defects.

One can speculate that the discrepancy between the behavior observed by Malekpour and coworkers and our study could be due to the nature of the defects considered. Indeed, in a very recent work Antidormi, Colombo and Roche carried out a systematic study of the thermal transport properties of amorphous graphene with varying structural quality, via MD simulations employing the optimized Tersoff potential [82]. Their work also quantifies the density of defects in terms of the deviation from an hexagonal lattice, similar to what we did in our amorphized samples. They find that the introduction of structural defects can reduce the thermal conductivity of graphene by more than two orders of magnitude, and for their highest defect concentration they obtain $\kappa \approx 10$ W/m-K, in good agreement with our results. Thus, the discrepancy between the studies mentioned above could indeed be due to the nature of the defects.

5.2. *Phagraphene: in-plane anisotropy in elastic modulus and thermal conductivity*

Some of the most remarkable electronic properties of graphene can be associated with the presence of Dirac cones in its electronic structure. In 2015, Wang and coworkers employed the first principles evolutionary algorithm USPEX to systematically search for 2D carbon allotropes [85]. During their search, graphene was found to have the lowest energy among the 2D carbon allotropes, and a new structure composed of 5-, 6- and 7-atom carbon rings, illustrated in Fig. 2(d), presented the second lowest energy. Wang and coworkers named the new carbon allotrope phagraphene, and their first principles calculations verified the presence of distorted Dirac cones in its electronic structure. The prediction of this novel 2D carbon allotrope caught our attention and led us to investigate its mechanical properties and thermal conductivity using MD simulations [83]. We described the C–C interactions with the Tersoff empirical potential with parameters optimized for graphene [56].

In order to verify the accuracy of the chosen potential in describing the atomic structure of phagraphene we calculated its phonon dispersion via lattice dynamics. The vibrational spectrum did not present any phonon branches with imaginary frequencies, indicating its structural stability. By performing uniaxial tensile simulations along both in-plane directions, we studied the deformation process and mechanical response of phagraphene. We found that phagraphene exhibits a remarkably high and direction-independent tensile strength around $\sigma^{3D} = 85 \pm 2$ GPa, whereas its elastic modulus is anisotropic along the in-plane directions, with values of $E^{3D} = 870 \pm 15$ GPa and $E^{3D} = 800 \pm 14$ GPa for the armchair and zigzag directions, respectively. Our simulations also indicated a different fracture behavior along both directions, with the armchair direction being more brittle and the zigzag direction more ductile, as shown in Fig. 2(e). Naturally, this in-plane anisotropy should also reflect in the thermal transport properties.

We performed NEMD simulations with cells of increasing length, and calculated their thermal conductivity along the armchair and zigzag directions independently. The intrinsic room temperature thermal conductivity of phagraphene along each in-plane direction was obtained from an extrapolation of the simulation results to systems of infinite length, as reproduced in Fig. 2(f). We found $\kappa = 218 \pm 20$ W/m-K along the armchair direction and $\kappa = 285 \pm 29$ W/m-K along the zigzag direction. These values are one order of magnitude smaller than the corresponding one for graphene $\kappa = 2900 \pm 100$ W/m-K obtained with the same empirical potential [76]. In order to better understand the difference between thermal conductivities of graphene and phagraphene, we calculated the phonon group velocities from the phonon dispersions of each material. Overall, we obtained an average group velocity of 780 m/s for graphene, while the equivalent average for phagraphene was only 125 m/s. This reduction in phonon group velocities is not the only reason for the lower thermal conductivity of phagraphene relative to graphene, but it provides further insight into the origin of the reduction.

5.3. *2D carbon ene-yne graphyne: highly stretchable with small thermal conductivity*

In 2017, Jia and coworkers reported the synthesis of a novel 2D carbon allotrope illustrated in Fig. 2(g), which they named carbon ene-yne (CEY) [86]. The novel material was synthesized from tetraethynylethene by solvent-phase reaction at low temperature and atmospheric pressure. First principles calculations revealed that CEY was a semiconductor with a small band gap of only 0.05 eV. Electron microscopy revealed a layered structure in CEY, with an interlayer spacing of 0.45 nm. Soon after the report by Jia and coworkers, we carried out an extensive computational evaluation of the mechanical, electronic, optical and thermal properties of 2D CEY [84]. We also performed first principles MD simulations to verify the stability of CEY monolayers up to 1500 K.

Investigating the mechanical response of CEY under uniaxial and biaxial strain, we obtained its stress–strain curves from DFT calculations. In the case of uniaxial strain, CEY presented a pronounced non-linear behavior, with a very small elastic modulus, ultimate tensile strength $\sigma^{2D} = 29$ GPa $\cdot$ nm, and ultimate strain of 0.30, as shown in Fig. 2(h). In the case of biaxial strain, we analyzed the stress along two perpendicular in-plane directions independently. The stress–strain relation along both directions is initially linear, followed by a non-linear regime up to the ultimate strain at 0.15. We also observed that the elastic modulus and the ultimate stress are anisotropic by a factor of two, assuming values around $E^{2D} = 100$ GPa $\cdot$ nm and $\sigma^{2D} = 7.5$ GPa $\cdot$ nm for the softer direction.

We then employed NEMD simulations based on the Tersoff potential with the parameter set optimized for carbon nanostructures to evaluate the lattice thermal conductivity of CEY monolayers along both in-plane directions. Fitting the simulation data with a phenomenological expression

$$\kappa(L) = \frac{\kappa}{1 + \Lambda/L}, \tag{9}$$

where L is the length of the system along the heat current direction and Λ is an effective phonon mean free path, we could obtain the intrinsic thermal conductivity of the material. We obtained very low room-temperature thermal conductivities of 10.2 ± 0.9 W/m-K and 3.44 ± 0.30 W/m-K, along the harder and softer directions respectively, as reproduced in Fig. 2(i). An analysis of phonon group velocities corroborated the anisotropy in the thermal conductivity, and also showed that along both directions the group velocities are one order of magnitude smaller than the corresponding value for graphene.

6. Mechanical properties and thermal conductivity of 2D B-C-N materials

6.1. Nitrogenated holey graphene: porous C_2N monolayers

One of the major challenges to the development of graphene-based electronic devices is the absence of a suitable band gap [87]. A possible mechanism to achieve a band gap would be to substitutionally dope graphene, however it still is very difficult to do so in a controllable fashion. In 2015, Mahmood and coworkers reported the synthesis of a layered two-dimensional structure with C_2N stoichiometry, which presented evenly distributed holes and nitrogen atoms in its basal plane as illustrated in Fig. 3(a) [88]. Those 2D structures, synthesized via a simple wet-chemical reaction, presented calculated and experimental band gaps of approximately 1.70 and 1.96 eV, respectively. Inspired by their experiments, we set to investigate the mechanical properties and the thermal conductivity of this novel material, dubbed nitrogenated holey graphene (NHG) [89].

We performed first principles calculations and MD simulations to investigate electronic, mechanical and heat transport properties of NHG at various temperatures. Although the Tersoff potential had not been parameterized for NHG specifically, parameters for graphene and hexagonal boron nitride (hBN) were available separately [56,90]. Therefore, we modeled C–C interactions with the available parameters and obtained the parameters for C–N via the mixing rules for the Tersoff potential. First principles calculations based on density functional theory yielded an elastic modulus $E^{3D} = 400\pm 5$ GPa at 0 K, 10% larger than predicted by MD simulations at low temperatures. We observed an overall decreasing trend in elastic modulus and tensile strength as temperature increases as shown in Fig. 3(b), again in accordance with general theory. At room temperature, we found that NHG can present a remarkable elastic modulus $E^{3D} = 335 \pm 5$ GPa and tensile strength $\sigma^{3D} = 60$ GPa.

We also investigated the thermal conductivity of NHG via NEMD simulations, and its size-dependence is shown in Fig. 3(c). At 300 K, an intrinsic thermal conductivity of 64.8 W/m-K was found, with an effective phonon mean free path of 34.0 nm, both of which are smaller than the respective values for graphene. Investigating the temperature dependence of the thermal conductivity, we predicted that it does not decay with the inverse temperature as expected for general materials above room temperature. We found that $\kappa \sim T^{-0.7}$, a behavior we attribute to the dimensionality of NHG, as well as to the presence of holes and two different atomic species in the material.

6.2. BC_3 and BC_6N: outstanding strength and thermal conductivity

Boron and nitrogen are two of the most common chemical elements found in carbon materials. Polyaniline carbon-nitride (C_3N) and boron-carbide (BC_3) are two examples that exhibit exceptional physical

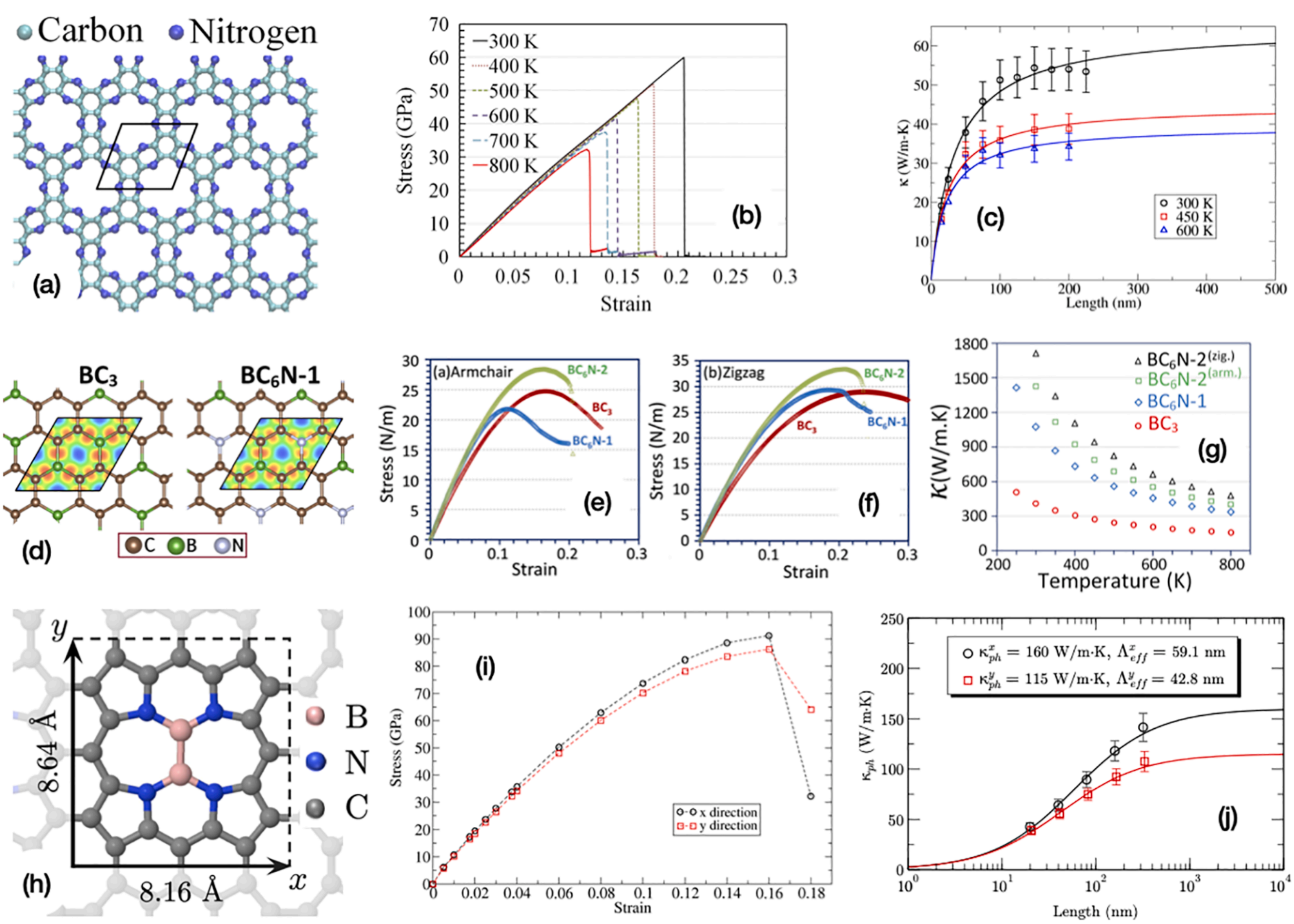

Fig. 3. (a) to (c) Atomic structure, temperature-dependent stress–strain curve and thermal conductivity of nitrogenated holey graphene. Adapted from Ref. [89]. (d) to (g) Atomic structure, stress–strain curve and thermal conductivity of BC_3 and BC_6N monolayers. Adapted from Ref. [91]. (h) to (j) Atomic structure, stress–strain curve and thermal conductivity of diboron-porphyrin monolayer. Adapted from Ref. [92].

properties [93–95]. Motivated by the versatility of BCN systems, we proposed to investigate two possible graphene-like materials with BC_6N stoichiometry, which differ only by the absence or presence of B-N bonds [91] In BC_6N-1, each C atom is bonded to a B or a N atom, besides another C atom, and B atoms never bond with N. Meanwhile, in BC_6N-2 each B atom in bonded to a N atom, such that some of the C atoms are not bonded to B nor N.

In this case, we employed exclusively first principles calculations to explore the structural stability, mechanical response, electronic, optical and thermal transport characteristics of BC_3 and BC_6N monolayers. The mechanical properties of BC_3 and BC_6N monolayers shown in Fig. 3(d) were investigated by uniaxial tensile simulations along armchair and zigzag in-plane directions, which allowed us to identify a possible anisotropy in the mechanical response. The lattice thermal conductivity of monolayer BC_3 and BC_6N was calculated by a fully iterative solution of the BTE, as implemented in the ShengBTE package [36].

Stress–strain curves show that the elastic properties of the monolayers are isotropic, and that both BC_6N structures have the same elastic moduli, which is a reasonable result since a small change in chemical environment should not have a large impact on the elastic properties. The elastic moduli of monolayer BC_3 and BC_6N were predicted to be considerably high at $E^{2D} = 256$ N/m and $E^{2D} = 305$ N/m, respectively, with the latter surpassing graphene ($E^{2D} = 300$ N/m) although a little lower than the result reported for C_3N ($E^{2D} = 340$ N/m [94]). BC_3 showed a small anisotropy in tensile strength with $\sigma^{2D} = 24.7$ N/m along the armchair direction and $\sigma^{2D} = 29.0$ N/m in the case of loading along the zigzag direction. This behavior is consistent with its more uniform atomic structure. In the case of BC_6N-1 monolayer we found $\sigma^{2D} = 21.8$ N/m and $\sigma^{2D} = 29.3$ N/m for uniaxial loading along armchair and zigzag directions, respectively. Meanwhile, BC_6N-2 exhibited the highest tensile strengths at $\sigma^{2D} = 28.4$ N/m and $\sigma^{2D} = 33.4$ N/m in the armchair and zigzag directions, respectively. The stress–strain curves are reproduced in Fig. 3(e) and (f).

Our calculations showed that the lattice thermal conductivity of BC_3 and BC_6N-1 are isotropic along the in-plane directions, while BC_6N-2 presented a clear anisotropy in its phonon conductivity. In the temperature interval considered, the conductivity of BC_6N-2 in either direction was always larger than that of BC_3 and BC_6N-1, as shown in Fig. 3(g). The room temperature lattice thermal conductivities are 410 W/m-K for BC_3, 1080 W/m-K in the case of BC_6N-1 (the structure without B-N bonds), 1430 and 1710 W/m-K along the armchair and zigzag directions of BC_6N-2. For comparison, we notice that the thermal conductivity of BC_6N monolayers is larger than C_3N, which was predicted to be 815 ± 20 W/m-K [94]. It is worth noticing that although both BCN structures present the same elastic modulus, BC_6N-2 presents the highest tensile strengths, which correlate with the highest thermal conductivities. Finally, it might seem counterintuitive for BC_6N-2 to present higher lattice thermal conductivities relative to BC_3 and BC_6N-1, but this behavior is explained by the higher phonon group velocities in BC_6N-2, particularly in the acoustic phonon range [91].

6.3. *Diboron-porphyrin monolayer: another 2D semiconductor with anisotropic elastic moduli and thermal conductivity*

In recent years there has been growing interest in 2D porphyrin-based polymers due to their possible application in catalysis [96], hydrogen storage [97], and spintronics [98]. Another advantage of porphyrin-based structures is the possibility of building large systems through a polymerization process. In 2020, we investigated a new porphyrin-based semiconductor, considering a unit cell consisting of a diboron-porphyrin molecule where the C–H bonds are replaced by C–C bonds, building a 2D supercell with $BC_{10}N_2$ stoichiometry, illustrated in Fig. 3(h) [92]. We performed first principles calculations to verify the mechanical stability of the material at 0 K as well as finite temperatures up to 4000 K. Further electronic structure calculations predicted a direct energy band gap of 0.6 eV. Thus, we employed NEMD simulations based on the Tersoff potential to obtain the lattice thermal conductivity of this novel material.

The stress–strain curves for diboron-porphyrin monolayers were obtained from first principles by uniaxial tensile calculations along both in-plane directions independently. Although these results were not actually reported in the manuscript, they are presented here in Fig. 3(i). We found a small anisotropy in room-temperature elastic moduli with values of $E^{3D} = 910 \pm 25$ GPa and $E^{3D} = 870 \pm 23$ GPa, along the x- and y- directions respectively, which correspond to zigzag and armchair in similar structures. The corresponding tensile strengths were $\sigma^{3D} = 91$ GPa and $\sigma^{3D} = 86$ GPa, again along zigzag and armchair directions respectively. The ultimate strain along both directions was estimated to be 0.17 ± 0.1. For the lattice thermal conductivities at room temperature we obtained $\kappa = 160$ W/m-K and $\kappa = 115$ W/m-K along the zigzag and armchair directions respectively, as shown in Fig. 3(j). Thus, the in-plane direction with larger elastic modulus and tensile strength also presents the largest thermal conductivity. It is interesting to notice that 2D diboron-porphyrin presents a high elastic modulus, only 15% smaller than pristine graphene, but a much lower thermal conductivity.

7. Summary, conclusions and perspectives

In Table 1 I summarize the elastic modulus, ultimate tensile strength, ultimate strain and lattice thermal conductivity of 2D carbon allotropes and B-C-N 2D materials investigated by us over the last few years. There is certainly some scatter in the data, but it is possible to notice a general trend in the relationship between lattice thermal conductivity and elastic modulus. In Fig. 4 I plot κ as a function of E for the data in Table 1, and include a dashed line representing the general data trend obtained from a linear regression and given by $\kappa = 1.12 e^{0.007E}$. The functional form for this relationship has no particular physical meaning, and was chosen empirically as the best fit to the data. A similar trend in the dependence between the thermal conductivity and the elastic modulus of entropy-stabilized oxides has been compiled recently by Braun and coworkers [99].

Let me conclude by asserting that computer simulations or computational experiments are a powerful approach to predict and understand materials properties, complementing experiments and theories.

Table 1
Room temperature elastic modulus, ultimate tensile strength, ultimate strain and lattice thermal conductivity of 2D carbon allotropes and B-C-N materials

Material	E^{3D} (GPa)	σ_U^{3D} (GPa)	ϵ_U	κ (W/m-K)	Ref.
Graphene	960 ± 50	132	0.21	2900 ± 100	[78,76]
Amorph. graphene (5%)	850	95	0.12	95	[78]
Amorph. graphene (35%)	500	50	0.16	15	[78]
Phagraphene (Arm.)	870 ± 15	85	0.13	218 ± 20	[83]
Phagraphene (Zig.)	800 ± 14	85	0.15	285 ± 29	[83]
CEY (Arm.)	222	7.5	0.15	3.44 ± 0.30	[84]
CEY (Zig.)	444	15	0.15	10.2 ± 0.9	[84]
NHG	335 ± 5	60	0.21	64.8	[89]
BC_3 (Arm.)	731	71	–	410	[91]
BC_3 (Zig.)	731	83	–	410	[91]
BC_6N-1 (Arm.)	871	62	–	1080	[91]
BC_6N-1 (Zig.)	871	84	–	1080	[91]
BC_6N-2 (Arm.)	871	81	–	1430	[91]
BC_6N-2 (Zig.)	871	95	–	1710	[91]
$BC_{10}N_2$ (x)	910 ± 25	91	0.17	160	[92]
$BC_{10}N_2$ (y)	870 ± 23	86	0.17	115	[92]

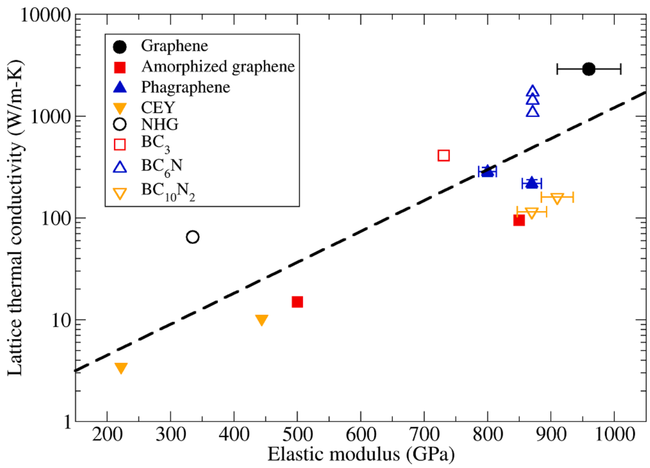

Fig. 4. Lattice thermal conductivity versus elastic modulus for 2D carbon allotropes (filled symbols) and B-C-N materials (empty symbols). The dashed line was obtained by linear regression to the data and represents a general trend given by $\kappa = 1.12e^{0.007E}$.

Molecular dynamics simulations can be used to calculate the mechanical properties and the lattice thermal conductivity of materials, in a range of temperatures and in the presence of defects, dopants and physico-chemical effects. Those results can be directly compared to experiments, but can also be used to probe the relationship between the elastic properties and the lattice thermal conductivity of materials, as shown in Fig. 4.

In the years ahead the use of machine learning techniques in materials modeling, sometimes called materials informatics, will become ever more important and will enhance the reach of the current computational methods [100–103]. Recent works have employed machine trained empirical potentials to model mechanical and heat transport properties of materials, achieving the precision of first principles methods at a fraction of the computational cost. In particular, moment tensor potentials can be trained by first principles MD simulations at a range of temperatures to reproduce interatomic forces and phonon properties of materials with high precision [104–110]. Finally, artificial intelligence techniques, such as deep leaning and neural networks are particularly suited to predict which materials properties have the larger influence on mechanical and heat transport characteristics.

CRediT authorship contribution statement

Luiz Felipe C. Pereira: Conceptualization, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization, Project administration, Funding acquisition.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

I would like to thank mentors and colleagues from whom I learned so much over the years, in particular F. G. Brady Moreira, David P. Landau, Mauro S. Ferreira, Davide Donadio, Bohayra Mortazavi, Zheyong Fan and Ari Harju. My research work has been funded by the Max Planck Society, Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES, Grant 88887.124762/2014-00) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq, Grants 309961/2017, 436859/2018 and 313462/2020). I am also grateful for the provision of computational resources by Rechenzentrum Garching of the Max Planck Society, the Jülich Supercomputing Centre under NIC project HMZ26, and the High Performance Computing Center (NPAD) at Universidade Federal do Rio Grande do Norte.

References

[1] H.W. Kroto, J.R. Heath, S.C. O'Brien, R.F. Curl, R.E. Smalley, C60: Buckminsterfullerene, Nature 318 (1985) 162.
[2] S. Iijima, Helical microtubules of graphitic carbon, Nature 354 (1991) 56–58.
[3] K.S. Novoselov, A.K. Geim, S.V. Morozov, D. Jiang, Y. Zhang, S.V. Dubonos, I. V. Grigorieva, A.A. Firsov, Electric field effect in atomically thin carbon films, Science 306 (2004) 666, https://doi.org/10.1126/science.1102896.
[4] A.K. Geim, K.S. Novoselov, The rise of graphene, Nat. Mater. 6 (3) (2007) 183, https://doi.org/10.1038/nmat1849.
[5] A.H. Castro Neto, F. Guinea, N.M.R. Peres, K.S. Novoselov, A.K. Geim, The electronic properties of graphene, Rev. Mod. Phys. 81 (1) (2009) 109–162, https://doi.org/10.1103/RevModPhys.81.109.
[6] K.S. Novoselov, D. Jiang, F. Schedin, T.J. Booth, V.V. Khotkevich, S.V. Morozov, A.K. Geim, Two-dimensional atomic crystals, Proc. Natl. Acad. Sci. 102 (30) (2005) 10451–10453, https://doi.org/10.1073/pnas.0502848102. http://www.pnas.org/cgi/doi/10.1073/pnas.0502848102.
[7] Y. Hernandez, V. Nicolosi, M. Lotya, F.M. Blighe, Z. Sun, S. De, I.T. McGovern, B. Holland, M. Byrne, Y.K. Gun'Ko, J.J. Boland, P. Niraj, G. Duesberg, S. Krishnamurthy, R. Goodhue, J. Hutchison, V. Scardaci, A.C. Ferrari, J. N. Coleman, High-yield production of graphene by liquid-phase exfoliation of graphite, Nat. Nanotechnol. 3 (9) (2008) 563–568, https://doi.org/10.1038/nnano.2008.215.
[8] J.N. Coleman, M. Lotya, A. O'Neill, S.D. Bergin, P.J. King, U. Khan, K. Young, A. Gaucher, S. De, R.J. Smith, I.V. Shvets, S.K. Arora, G. Stanton, H.-Y. Kim, K. Lee, G.T. Kim, G.S. Duesberg, T. Hallam, J.J. Boland, J.J. Wang, J.F. Donegan, J.C. Grunlan, G. Moriarty, A. Shmeliov, R.J. Nicholls, J.M. Perkins, E.M. Grieveson, K. Theuwissen, D.W. McComb, P.D. Nellist, V. Nicolosi, Two-Dimensional Nanosheets Produced by Liquid Exfoliation of Layered Materials, Science (80-.). 331 (6017) (2011) 568–571. doi:10.1126/science.1194975. http://www.sciencemag.org/cgi/doi/10.1126/science.1194975.
[9] X. Li, W. Cai, J. An, S. Kim, J. Nah, D. Yang, R. Piner, A. Velamakanni, I. Jung, E. Tutuc, S.K. Banerjee, L. Colombo, R.S. Ruoff, Large-Area Synthesis of High-Quality and Uniform Graphene Films on Copper Foils, Science (80-.) 324 (5932) (2009) 1312–1314, https://doi.org/10.1126/science.1171245.
[10] D.G. Papageorgiou, I.A. Kinloch, R.J. Young, Mechanical properties of graphene and graphene-based nanocomposites, Prog. Mater. Sci. 90 (2017) 75–127, https://doi.org/10.1016/j.pmatsci.2017.07.004 https://linkinghub.elsevier.com/retrieve/pii/S0079642517300968.
[11] A.A. Balandin, S. Ghosh, D.L. Nika, E.P. Pokatilov, Thermal Conduction in Suspended Graphene Layers, Fullerenes, Nanotub. Carbon Nanostruct. 18 (4–6) (2010) 474–486, https://doi.org/10.1080/1536383X.2010.487785. http://www.tandfonline.com/doi/abs/10.1080/1536383X.2010.487785.
[12] A.A. Balandin, Thermal properties of graphene and nanostructured carbon materials, Nat. Mater. 10 (8) (2011) 569–581, https://doi.org/10.1038/nmat3064. http://www.nature.com/doifinder/10.1038/nmat3064.
[13] D.L. Nika, A.A. Balandin, Phonons and thermal transport in graphene and graphene-based materials, Reports Prog. Phys. 80 (3) (2017), https://doi.org/10.1088/1361-6633/80/3/036502. https://iopscience.iop.org/article/10.1088/1361-6633/80/3/036502, 036502.
[14] A.A. Balandin, Phononics of Graphene and Related Materials, ACS Nano 14 (5) (2020) 5170–5178, https://doi.org/10.1021/acsnano.0c02718. https://pubs.acs.org/doi/10.1021/acsnano.0c02718.
[15] N. Metropolis, A.W. Rosenbluth, M.N. Rosenbluth, A.H. Teller, E. Teller, Equation of state calculations by fast computing machines, J. Chem. Phys. 21 (6) (1953) 1087–1092, https://doi.org/10.1063/1.1699114. http://aip.scitation.org/doi/10.1063/1.1699114.
[16] E. Fermi, J. Pasta, S. Ulam, Studies of nonlinear problems (1955). doi:AC11-98PN38206. http://www.osti.gov/accomplishments/pdf/A80037041/01.pdf.
[17] M.P. Allen, D.J. Tildesley, Computer Simulation of Liquids, Oxford University Press, 1989.
[18] R. Car, M. Parrinello, Unified approach for molecular dynamics and density-functional theory, Phys. Rev. Lett. 55 (22) (1985) 2471–2474, https://doi.org/10.1103/PhysRevLett.55.2471. http://link.aps.org/doi/10.1103/PhysRevLett.55.2471 https://link.aps.org/doi/10.1103/PhysRevLett.55.2471.
[19] H. Huntington, The elastic constants of crystals, London, Edinburgh, Dublin Philos. Mag. J. Sci. 45 (269) (1958) 213–351, https://doi.org/10.1016/S0081-1947(08)60553-6. https://www.tandfonline.com/doi/full/10.1080/14786442308636317 https://linkinghub.elsevier.com/retrieve/pii/S0081194708605536.
[20] D. Akinwande, C.J. Brennan, J.S. Bunch, P. Egberts, J.R. Felts, H. Gao, R. Huang, J.-S. Kim, T. Li, Y. Li, K.M. Liechti, N. Lu, H.S. Park, E.J. Reed, P. Wang, B. I. Yakobson, T. Zhang, Y.-W. Zhang, Y. Zhou, Y. Zhu, A review on mechanics and mechanical properties of 2D materials Graphene and beyond, Extrem. Mech. Lett. 13 (2017) 42–77, https://doi.org/10.1016/j.eml.2017.01.008. https://linkinghub.elsevier.com/retrieve/pii/S235243161630236X.
[21] M.S. Green, Markoff random processes and the statistical mechanics of time-dependent phenomena. II. Irreversible processes in fluids, J. Chem. Phys. 22 (3) (1954) 398, https://doi.org/10.1063/1.1740082.

[22] R. Kubo, Statistical-mechanical theory of irreversible processes. I. General theory and simple applications to magnetic and conduction problems, J. Phys. Soc. Jpn. 12 (6) (1957) 570–586, https://doi.org/10.1143/JPSJ.12.570.

[23] A. Marcolongo, P. Umari, S. Baroni, Microscopic theory and quantum simulation of atomic heat transport, Nat. Phys. 12 (1) (2016) 80–84, https://doi.org/10.1038/nphys3509. http://www.nature.com/articles/nphys3509.

[24] L. Isaeva, G. Barbalinardo, D. Donadio, S. Baroni, Modeling heat transport in crystals and glasses from a unified lattice-dynamical approach, Nat. Commun. 10 (1) (2019) 3853, https://doi.org/10.1038/s41467-019-11572-4. http://www.nature.com/articles/s41467-019-11572-4.

[25] C. Carbogno, R. Ramprasad, M. Scheffler, Ab initio Green-Kubo approach for the thermal conductivity of solids, Phys. Rev. Lett. 118 (17) (2017), https://doi.org/10.1103/PhysRevLett.118.175901. http://link.aps.org/doi/10.1103/PhysRevLett.118.175901, 175901.

[26] M. Simoncelli, N. Marzari, F. Mauri, Unified theory of thermal transport in crystals and glasses, Nat. Phys. 15 (8) (2019) 809–813, https://doi.org/10.1038/s41567-019-0520-x. http://www.nature.com/articles/s41567-019-0520-x.

[27] E. Helfand, Transport coefficients from dissipation in a canonical ensemble, Phys. Rev. 119 (1) (1960) 1–9, https://doi.org/10.1103/PhysRev.119.1. http://link.aps.org/doi/10.1103/PhysRev.119.1.

[28] F. Müller-Plathe, A simple nonequilibrium molecular dynamics method for calculating the thermal conductivity, J. Chem. Phys. 106 (14) (1997) 6082–6085, https://doi.org/10.1063/1.473271. http://aip.scitation.org/doi/10.1063/1.473271.

[29] E. Lampin, P.L. Palla, P.-A. Francioso, F. Cleri, Thermal conductivity from approach-to-equilibrium molecular dynamics, J. Appl. Phys. 114 (3) (2013), https://doi.org/10.1063/1.4815945. http://aip.scitation.org/doi/10.1063/1.4815945, 033525.

[30] C. Melis, R. Dettori, S. Vandermeulen, L. Colombo, Calculating thermal conductivity in a transient conduction regime: theory and implementation, Eur. Phys. J. B 87 (4) (2014) 96, https://doi.org/10.1140/epjb/e2014-50119-0. http://link.springer.com/10.1140/epjb/e2014-50119-0.

[31] J. Ziman, Electrons and Phonons, Oxford University Press, Oxford, UK, 1960.

[32] M.T. Dove, Introduction to Lattice Dynamics, Cambridge University Press, Cambridge, UK, 1993.

[33] L. Lindsay, D.A. Broido, N. Mingo, Lattice thermal conductivity of single-walled carbon nanotubes: Beyond the relaxation time approximation and phonon-phonon scattering selection rules, Phys. Rev. B 80 (12) (2009), https://doi.org/10.1103/PhysRevB.80.125407. http://link.aps.org/doi/10.1103/PhysRevB.80.125407, 125407.

[34] G. Fugallo, M. Lazzeri, L. Paulatto, F. Mauri, Ab initio variational approach for evaluating lattice thermal conductivity, Phys. Rev. B 88 (4) (2013), https://doi.org/10.1103/PhysRevB.88.045430. https://link.aps.org/doi/10.1103/PhysRevB.88.045430, 045430.

[35] G. Fugallo, A. Cepellotti, L. Paulatto, M. Lazzeri, N. Marzari, F. Mauri, Thermal conductivity of graphene and graphite: collective excitations and mean free paths, Nano Lett. 14 (11) (2014) 6109–6114, https://doi.org/10.1021/nl502059f.

[36] W. Li, J. Carrete, N.A. Katcho, N. Mingo, ShengBTE: A solver of the Boltzmann transport equation for phonons, Comp. Phys. Comm. 185 (6) (2014) 1747–1758, https://doi.org/10.1016/j.cpc.2014.02.015. http://linkinghub.elsevier.com/retrieve/pii/S0010465514000484.

[37] A. Cepellotti, G. Fugallo, L. Paulatto, M. Lazzeri, F. Mauri, N. Marzari, Phonon hydrodynamics in two-dimensional materials, Nat. Commun. 6 (2015) 6400, https://doi.org/10.1038/ncomms7400. http://www.nature.com/doifinder/10.1038/ncomms7400.

[38] A.J.H. McGaughey, A. Jain, H.-Y. Kim, Phonon properties and thermal conductivity from first principles, lattice dynamics, and the Boltzmann transport equation, J. Appl. Phys. 125 (1) (2019), https://doi.org/10.1063/1.5064602. http://aip.scitation.org/doi/10.1063/1.5064602, 011101.

[39] A. Debernardi, S. Baroni, E. Molinari, Anharmonic phonon lifetimes in semiconductors from density-functional perturbation theory, Phys. Rev. Lett. 75 (9) (1995) 1819, https://doi.org/10.1103/PhysRevLett.75.1819. http://link.aps.org/doi/10.1103/PhysRevLett.75.1819.

[40] J.E. Turney, E.S. Landry, A.J.H. McGaughey, C.H. Amon, Predicting phonon properties and thermal conductivity from anharmonic lattice dynamics calculations and molecular dynamics simulations, Phys. Rev. B 79 (6) (2009), https://doi.org/10.1103/PhysRevB.79.064301. http://link.aps.org/doi/10.1103/PhysRevB.79.064301, 064301.

[41] H. Dong, Z. Fan, L. Shi, A. Harju, T. Ala-Nissila, Equivalence of the equilibrium and the nonequilibrium molecular dynamics methods for thermal conductivity calculations: From bulk to nanowire silicon, Phys. Rev. B 97 (9) (2018), https://doi.org/10.1103/PhysRevB.97.094305. https://link.aps.org/doi/10.1103/PhysRevB.97.094305, 094305.

[42] M. Puligheddu, Y. Xia, M. Chan, G. Galli, Computational prediction of lattice thermal conductivity: A comparison of molecular dynamics and Boltzmann transport approaches, Phys. Rev. Mater. 3 (8) (2019), https://doi.org/10.1103/PhysRevMaterials.3.085401. https://link.aps.org/doi/10.1103/PhysRevMaterials.3.085401, 085401.

[43] Y. Hu, T. Feng, X. Gu, Z. Fan, X. Wang, M. Lundstrom, S.S. Shrestha, H. Bao, Unification of nonequilibrium molecular dynamics and the mode-resolved phonon Boltzmann equation for thermal transport simulations, Phys. Rev. B 101 (15) (2020), https://doi.org/10.1103/PhysRevB.101.155308. https://link.aps.org/doi/10.1103/PhysRevB.101.155308, 155308.

[44] G. Fugallo, L. Colombo, Calculating lattice thermal conductivity: a synopsis, Phys. Scr. 93 (4) (2018), https://doi.org/10.1088/1402-4896/aaa6f3. http://stacks.iop.org/1402-4896/93/i=4/a=043002?key=crossref.564f54974f76f9d7aabe1ee10cecd108, 043002.

[45] C. Lee, X. Wei, J.W. Kysar, J. Hone, Measurement of the elastic properties and intrinsic strength of monolayer graphene, Science (80-.). 321 (5887) (2008) 385–388. arXiv:47749150628, doi:10.1126/science.1157996.

[46] O.L. Blakslee, D.G. Proctor, E.J. Seldin, G.B. Spence, T. Weng, Elastic constants of compression annealed pyrolytic graphite, J. Appl. Phys. 41 (8) (1970) 3373–3382, https://doi.org/10.1063/1.1659428. http://aip.scitation.org/doi/10.1063/1.1659428.

[47] E. Hernández, C. Goze, P. Bernier, A. Rubio, Elastic properties of C and BxCyNz composite nanotubes, Phys. Rev. Lett. 80 (20) (1998) 4502–4505. http://link.aps.org/doi/10.1103/PhysRevLett.80.4502.

[48] K.N. Kudin, G.E. Scuseria, B.I. Yakobson, C2F, BN, and C nanoshell elasticity from ab initio computations, Phys. Rev. B 64 (23) (2001), https://doi.org/10.1103/PhysRevB.64.235406. https://link.aps.org/doi/10.1103/PhysRevB.64.235406, 235406.

[49] F. Liu, P. Ming, J. Li, Ab initio calculation of ideal strength and phonon instability of graphene under tension, Phys. Rev. B 76 (6) (2007), https://doi.org/10.1103/PhysRevB.76.064120. http://link.aps.org/doi/10.1103/PhysRevB.76.064120, 064120.

[50] E. Cadelano, P.L. Palla, S. Giordano, L. Colombo, Nonlinear elasticity of monolayer graphene, Phys. Rev. Lett. 102 (23) (2009), 235502, https://doi.org/10.1103/PhysRevLett.102.235502. https://link.aps.org/doi/10.1103/PhysRevLett.102.235502.

[51] D.W. Brenner, O.A. Shenderova, J.A. Harrison, S.J. Stuart, B. Ni, S.B. Sinnott, A second-generation reactive empirical bond order (REBO) potential energy expression for hydrocarbons, J. Phys. Condens. Matter 14 (4) (2002) 783–802, https://doi.org/10.1088/0953-8984/14/4/312. http://iopscience.iop.org/0953-8984/14/4/312 http://stacks.iop.org/0953-8984/14/i=4/a=312?key=crossref.563ba867ec801ce55fc736b3f8968942.

[52] J.-W. Jiang, J.-S. Wang, B. Li, Young's modulus of graphene: A molecular dynamics study, Phys. Rev. B 80 (11) (2009), https://doi.org/10.1103/PhysRevB.80.113405. http://link.aps.org/doi/10.1103/PhysRevB.80.113405, 113405.

[53] B. Mortazavi, S. Ahzi, V. Toniazzo, Y. Rémond, Nitrogen doping and vacancy effects on the mechanical properties of graphene: A molecular dynamics study, Phys. Lett. A 376 (12–13) (2012) 1146–1153, https://doi.org/10.1016/j.physleta.2011.11.034 https://linkinghub.elsevier.com/retrieve/pii/S0375960111014174.

[54] J. Tersoff, Empirical interatomic potential for carbon, with applications to amorphous carbon, Phys. Rev. Lett. 61 (25) (1988) 2879, http://link.aps.org/doi/10.1103/PhysRevLett.61.2879.

[55] J. Tersoff, New empirical approach for the structure and energy of covalent systems, Phys. Rev. B 37 (12) (1988) 6991–7000, https://doi.org/10.1103/PhysRevB.37.6991. http://link.aps.org/doi/10.1103/PhysRevB.37.6991.

[56] L. Lindsay, D.A. Broido, Optimized Tersoff and Brenner empirical potential parameters for lattice dynamics and phonon thermal transport in carbon nanotubes and graphene, Phys. Rev. B 81 (20) (2010), https://doi.org/10.1103/PhysRevB.81.205441. http://link.aps.org/doi/10.1103/PhysRevB.81.205441, 205441.

[57] A.A. Balandin, S. Ghosh, W. Bao, I. Calizo, D. Teweldebrhan, F. Miao, C.N. Lau, Superior thermal conductivity of single-layer graphene, Nano Lett. 8 (3) (2008) 902, https://doi.org/10.1021/nl0731872.

[58] W. Cai, A.L. Moore, Y. Zhu, X. Li, S. Chen, L. Shi, R.S. Ruoff, Thermal transport in suspended and supported monolayer graphene grown by chemical vapor deposition, Nano Lett. 10 (5) (2010) 1645–1651, https://doi.org/10.1021/nl9041966.

[59] L.A. Jauregui, Y. Yue, A.N. Sidorov, J. Hu, Q. Yu, G. Lopez, R. Jalilian, D. K. Benjamin, D.A. Delkd, W. Wu, Z. Liu, X. Wang, Z. Jiang, X. Ruan, J. Bao, S. S. Pei, Y.P. Chen, Thermal Transport in Graphene Nanostructures: Experiments and Simulations, ECS Trans. 28 (5) (2010) 73–83, https://doi.org/10.1149/1.3367938. http://link.aip.org/link/ECSTF8/v28/i5/p73/s1&Agg=doi.

[60] P.G. Klemens, D.F. Pedraza, Thermal conductivity of graphite in the basal plane, Carbon N.Y. 32 (4) (1994) 735–741, https://doi.org/10.1016/0008-6223(94)90096-5.

[61] D. Nika, E. Pokatilov, A. Askerov, A.A. Balandin, Phonon thermal conduction in graphene: Role of Umklapp and edge roughness scattering, Phys. Rev. B 79 (15) (2009), https://doi.org/10.1103/PhysRevB.79.155413. http://link.aps.org/doi/10.1103/PhysRevB.79.155413, 155413.

[62] B.D. Kong, S. Paul, M.B. Nardelli, K.W. Kim, First-principles analysis of lattice thermal conductivity in monolayer and bilayer graphene, Phys. Rev. B 80 (3)

(2009), https://doi.org/10.1103/PhysRevB.80.033406. http://link.aps.org/doi/10.1103/PhysRevB.80.033406, 033406.
[63] J. Tersoff, Structural properties of sp3-bonded hydrogenated amorphous carbon, Phys. Rev. B 44 (21) (1991) 12039–12042, https://doi.org/10.1103/PhysRevB.44.12039. http://link.aps.org/doi/10.1103/PhysRevB.44.12039.
[64] W.J. Evans, L. Hu, P. Keblinski, Thermal conductivity of graphene ribbons from equilibrium molecular dynamics: Effect of ribbon width, edge roughness, and hydrogen termination, Appl. Phys. Lett. 96 (20) (2010), https://doi.org/10.1063/1.3435465. http://link.aip.org/link/APPLAB/v96/i20/p203112/s1&Agg=doi, 203112.
[65] J.A. Thomas, R.M. Iutzi, A.J.H. McGaughey, Thermal conductivity and phonon transport in empty and water-filled carbon nanotubes, Phys. Rev. B 81 (4) (2010), https://doi.org/10.1103/PhysRevB.81.045413. http://link.aps.org/doi/10.1103/PhysRevB.81.045413, 045413.
[66] A. Lippi, R. Livi, Heat conduction in two-dimensional nonlinear lattices, J. Stat. Phys. 100 (2000) 1147–1172, http://www.springerlink.com/index/MH60X378752GV73M.pdf.
[67] S. Lepri, R. Livi, A. Politi, Thermal conduction in classical low-dimensional lattices, Phys. Rep. 377 (1) (2003) 1–80, https://doi.org/10.1016/S0370-1573(02)00558-6. http://linkinghub.elsevier.com/retrieve/pii/S0370157302005586.
[68] S. Lepri, R. Livi, A. Politi, Studies of thermal conductivity in Fermi-Pasta-Ulam-like lattices, Chaos 15 (1) (2005) 15118, https://doi.org/10.1063/1.1854281.
[69] L. Wang, B. Hu, B. Li, Logarithmic divergent thermal conductivity in two-dimensional nonlinear lattices, Phys. Rev. E 86 (4) (2012), https://doi.org/10.1103/PhysRevE.86.040101. http://link.aps.org/doi/10.1103/PhysRevE.86.040101, 040101.
[70] S. Liu, X.F. Xu, R.G. Xie, G. Zhang, B.W. Li, Anomalous heat conduction and anomalous diffusion in low dimensional nanoscale systems, Eur. Phys. J. B 85 (10) (2012) 337, https://doi.org/10.1140/epjb/e2012-30383-8.
[71] S. Plimpton, Fast parallel algorithms for short-range molecular dynamics, J. Comput. Phys. 117 (1) (1995) 1–19, https://doi.org/10.1006/jcph.1995.1039.
[72] L.F.C. Pereira, D. Donadio, Divergence of the thermal conductivity in uniaxially strained graphene, Phys. Rev. B 87 (12) (2013), 125424, https://doi.org/10.1103/PhysRevB.87.125424. http://link.aps.org/doi/10.1103/PhysRevB.87.125424.
[73] X. Xu, L.F.C. Pereira, Y. Wang, J. Wu, K. Zhang, X. Zhao, S. Bae, C. Tinh Bui, R. Xie, J.T.L. Thong, B.H. Hong, K.P. Loh, D. Donadio, B. Li, B. Özyilmaz, Length-dependent thermal conductivity in suspended single-layer graphene, Nat. Commun. 5 (2014) 3689, https://doi.org/10.1038/ncomms4689.
[74] G. Barbarino, C. Melis, L. Colombo, Intrinsic thermal conductivity in monolayer graphene is ultimately upper limited: A direct estimation by atomistic simulations, Phys. Rev. B 91 (3) (2015), https://doi.org/10.1103/PhysRevB.91.035416, 035416.
[75] Z. Fan, L.F.C. Pereira, H.-Q. Wang, J.-C. Zheng, D. Donadio, A. Harju, Force and heat current formulas for many-body potentials in molecular dynamics simulations with applications to thermal conductivity calculations, Phys. Rev. B 92 (9) (2015), https://doi.org/10.1103/PhysRevB.92.094301, 094301.
[76] Z. Fan, L.F.C. Pereira, P. Hirvonen, M.M. Ervasti, K.R. Elder, D. Donadio, T. Ala-Nissila, A. Harju, Thermal conductivity decomposition in two-dimensional materials: Application to graphene, Phys. Rev. B 95 (14) (2017), 144309, https://doi.org/10.1103/PhysRevB.95.144309.
[77] J. Kotakoski, C. Brand, Y. Lilach, O. Cheshnovsky, C. Mangler, M. Arndt, J. C. Meyer, Toward two-dimensional all-carbon heterostructures via ion beam patterning of single-layer graphene, Nano Lett. 15 (9) (2015) 5944–5949, https://doi.org/10.1021/acs.nanolett.5b02063.
[78] B. Mortazavi, Z. Fan, L.F.C. Pereira, A. Harju, T. Rabczuk, Amorphized graphene: A stiff material with low thermal conductivity, Carbon N.Y. 103 (2016) 318, https://doi.org/10.1016/j.carbon.2016.03.007, http://linkinghub.elsevier.com/retrieve/pii/S0008622316301968.
[79] B. Mortazavi, T. Rabczuk, Multiscale modeling of heat conduction in graphene laminates, Carbon N.Y. 85 (2015) 1–7, https://doi.org/10.1016/j.carbon.2014.12.046, http://linkinghub.elsevier.com/retrieve/pii/S000862231401207X.
[80] A. Bagri, S.-p. Kim, R.S. Ruoff, V.B. Shenoy, Thermal transport across twin grain boundaries in polycrystalline graphene from nonequilibrium molecular dynamics simulations, Nano Lett 11 (9) (2011) 3917–3921, https://doi.org/10.1021/nl202118d.
[81] H. Malekpour, P. Ramnani, S. Srinivasan, G. Balasubramanian, D.L. Nika, A. Mulchandani, R.K. Lake, A.A. Balandin, Thermal conductivity of graphene with defects induced by electron beam irradiation, Nanoscale 8 (30) (2016) 14608–14616, https://doi.org/10.1039/C6NR03470E.
[82] A. Antidormi, L. Colombo, S. Roche, Thermal transport in amorphous graphene with varying structural quality, 2D Mater 8 (1) (2020), 015028, https://doi.org/10.1088/2053-1583/abc7f8.
[83] L.F.C. Pereira, B. Mortazavi, M. Makaremi, T. Rabczuk, Anisotropic thermal conductivity and mechanical properties of phagraphene: A molecular dynamics study, RSC Adv. 6 (2016) 57773, https://doi.org/10.1039/C6RA05082D.
[84] B. Mortazavi, M. Shahrokhi, T. Rabczuk, L.F.C. Pereira, Electronic, optical and thermal properties of highly stretchable 2D carbon Ene-yne graphyne, Carbon N. Y. 123 (2017) 344, https://doi.org/10.1016/j.carbon.2017.07.066.
[85] Z. Wang, X.-F. Zhou, X. Zhang, Q. Zhu, H. Dong, M. Zhao, A.R. Oganov, Phagraphene: A low-energy graphene allotrope composed of 5–6-7 carbon rings with distorted dirac cones, Nano Lett. 15 (9) (2015) 6182–6186, https://doi.org/10.1021/acs.nanolett.5b02512.
[86] Z. Jia, Z. Zuo, Y. Yi, H. Liu, D. Li, Y. Li, Y. Li, Low temperature, atmospheric pressure for synthesis of a new carbon Ene-yne and application in Li storage, Nano Energy 33 (2017) 343–349, https://doi.org/10.1016/j.nanoen.2017.01.049.
[87] J.-C. Charlier, S. Roche, Electronic and transport properties of nanotubes, Rev. Mod. Phys. 79 (2) (2007) 677–732, https://doi.org/10.1103/RevModPhys.79.677.
[88] J. Mahmood, E.K. Lee, M. Jung, D. Shin, I.-Y. Jeon, S.-M. Jung, H.-J. Choi, J.-M. Seo, S.-Y. Bae, S.-D. Sohn, N. Park, J.H. Oh, H.-J. Shin, J.-B. Baek, Nitrogenated holey two-dimensional structures, Nat. Commun. 6 (2015) 6486, https://doi.org/10.1038/ncomms7486.
[89] B. Mortazavi, O. Rahaman, T. Rabczuk, L.F.C. Pereira, Thermal conductivity and mechanical properties of nitrogenated holey graphene, Carbon N.Y. 106 (2016) 1. doi:10.1016/j.carbon.2016.05.009. http://linkinghub.elsevier.com/retrieve/pii/S0008622316303591.
[90] L. Lindsay, D.A. Broido, Enhanced thermal conductivity and isotope effect in single-layer hexagonal boron nitride, Phys. Rev. B 84 (15) (2011), https://doi.org/10.1103/PhysRevB.84.155421, 155421.
[91] B. Mortazavi, M. Shahrokhi, M. Raeisi, X. Zhuang, L.F.C. Pereira, T. Rabczuk, Outstanding strength, optical characteristics and thermal conductivity of graphene-like BC3 and BC6N semiconductors, Carbon N.Y. 149 (2019) 733–742. doi:10.1016/j.carbon.2019.04.084. https://linkinghub.elsevier.com/retrieve/pii/S0008622319304245.
[92] R.M. Tromer, I.M. Felix, A. Freitas, S. Azevedo, L.F.C. Pereira, Diboron-porphyrin monolayer: A new 2D semiconductor, Comput. Mater. Sci. 172 (2020), https://doi.org/10.1016/j.commatsci.2019.109338, 109338.
[93] S. Azevedo, R. de Paiva, Structural stability and electronic properties of carbon-boron nitride compounds, Europhys. Lett. 75 (1) (2006) 126–132, https://doi.org/10.1209/epl/i2006-10066-0.
[94] B. Mortazavi, Ultra high stiffness and thermal conductivity of graphene like C 3 N, Carbon N.Y. 118 (2017) 25–34. doi:10.1016/j.carbon.2017.03.029. http://linkinghub.elsevier.com/retrieve/pii/S0008622317302701.
[95] R.K. Zahedi, A. Shirazi, P. Alimouri, N. Alajlan, T. Rabczuk, Mechanical properties of graphene-like BC3; a molecular dynamics study, Comput. Mater. Sci. 168 (May) (2019) 1–10, https://doi.org/10.1016/j.commatsci.2019.05.053.
[96] M.E. Ali, B. Sanyal, P.M. Oppeneer, Tuning the magnetic interaction between manganese porphyrins and ferromagnetic co substrate through dedicated control of the adsorption, J. Phys. Chem. C 113 (32) (2009) 14381–14383, https://doi.org/10.1021/jp902644q.
[97] D.H. Lee, W.J. Lee, W.J. Lee, S.O. Kim, Y.-H. Kim, Theory, synthesis, and oxygen reduction catalysis of fe-porphyrin-like carbon nanotube, Phys. Rev. Lett. 106 (17) (2011), https://doi.org/10.1103/PhysRevLett.106.175502, 175502.
[98] W.J. Cho, Y. Cho, S.K. Min, W.Y. Kim, K.S. Kim, Chromium porphyrin arrays as spintronic devices, J. Am. Chem. Soc. 133 (24) (2011) 9364–9369, https://doi.org/10.1021/ja111565w.
[99] J.L. Braun, C.M. Rost, M. Lim, A. Giri, D.H. Olson, G.N. Kotsonis, G. Stan, D. W. Brenner, J.-P. Maria, P.E. Hopkins, Charge-Induced Disorder Controls the Thermal Conductivity of Entropy-Stabilized Oxides, Adv. Mater. 30 (51) (2018) 1805004, https://doi.org/10.1002/adma.201805004.
[100] G.R. Schleder, A.C.M. Padilha, C.M. Acosta, M. Costa, A. Fazzio, From DFT to machine learning: recent approaches to materials science-a review, J. Phys. Mater. 2 (3) (2019), https://doi.org/10.1088/2515-7639/ab084b, 032001.
[101] J. Schmidt, M.R. Marques, S. Botti, M.A. Marques, Recent advances and applications of machine learning in solid-state materials science, npj Comput, Mater 5 (1) (2019), https://doi.org/10.1038/s41524-019-0221-0.
[102] L. Chen, H. Tran, R. Batra, C. Kim, R. Ramprasad, Machine learning models for the lattice thermal conductivity prediction of inorganic materials, Comput. Mater. Sci. 170 (2019), https://doi.org/10.1016/j.commatsci.2019.109155, 109155.
[103] Y. Iwasaki, I. Takeuchi, V. Stanev, A.G. Kusne, M. Ishida, A. Kirihara, K. Ihara, R. Sawada, K. Terashima, H. Someya, K.-I. Uchida, E. Saitoh, S. Yorozu, Machine-learning guided discovery of a new thermoelectric material, Sci. Rep. 9 (1) (2019) 2751, https://doi.org/10.1038/s41598-019-39278-z.
[104] A.V. Shapeev, Moment Tensor Potentials: A class of systematically improvable interatomic potentials, Multiscale Model. Simul. 14 (3) (2016) 1153–1173. doi:10.1137/15M1054183. http://epubs.siam.org/doi/10.1137/15M1054183.
[105] K.T. Schütt, P.-J. Kindermans, H.E. Sauceda, S. Chmiela, A. Tkatchenko, K.-R. SchNet Müller, A continuous-filter convolutional neural network for modeling quantum interactions, Adv. Neural Inf. Process. Syst. 30 (2017) 992–1002. http://arxiv.org/abs/1706.08566.
[106] V. Botu, R. Batra, J. Chapman, R. Ramprasad, Machine learning force fields: construction, validation, and outlook, J. Phys. Chem. C 121 (1) (2017) 511–522, https://doi.org/10.1021/acs.jpcc.6b10908.

[107] Y. Zuo, C. Chen, X. Li, Z. Deng, Y. Chen, J. Behler, G. Csányi, A.V. Shapeev, A. P. Thompson, M.A. Wood, S.P. Ong, Performance and cost assessment of machine learning interatomic potentials, J. Phys. Chem. A 124 (4) (2020) 731–745, https://doi.org/10.1021/acs.jpca.9b08723.

[108] B. Mortazavi, I.S. Novikov, E.V. Podryabinkin, S. Roche, T. Rabczuk, A. V. Shapeev, X. Zhuang, Exploring phononic properties of two-dimensional materials using machine learning interatomic potentials, Appl. Mater. Today 20 (2020), https://doi.org/10.1016/j.apmt.2020.100685, 100685.

[109] B. Mortazavi, E.V. Podryabinkin, I.S. Novikov, T. Rabczuk, X. Zhuang, A. V. Shapeev, Accelerating first-principles estimation of thermal conductivity by machine-learning interatomic potentials: A MTP/ShengBTE solution, Comput. Phys. Commun. 258 (2021), https://doi.org/10.1016/j.cpc.2020.107583, 107583.

[110] B. Mortazavi, F. Shojaei, X. Zhuang, L.F.C. Pereira, First-principles investigation of electronic, optical, mechanical and heat transport properties of pentadiamond: A comparison with diamond, Carbon Trends 3 (2021), https://doi.org/10.1016/j.cartre.2021.100036, 100036.

Luiz Felipe C. Pereira obtained his B.Sc. and M.Sc. degrees in physics from Universidade Federal de Pernambuco in Recife, Brazil, and his Ph.D. from Trinity College Dublin, Ireland. He was a postdoctoral research fellow at the Max Planck Institute for Polymer Research in Mainz, Germany, and assistant professor at the Department of Theoretical and Experimental Physics of Universidade Federal do Rio Grande do Norte in Natal, Brazil. He is a research fellow of the Brazilian National Council for Scientific and Technological Development (CNPq), and an affiliated member of the Brazilian Academy of Sciences. Currently, he is a tenured assistant professor at the Department of Physics of Universidade Federal de Pernambuco in Recife, Brazil, where he leads a research group on computational materials science, collaborating with experimentalists and theoreticians worldwide.